Essential Steps for Implementing Natural Language Processing

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Introduction

- Using smart assistants (Amazon Alexa, Google Home, Apple Siri)
- Communicating with these assistants in natural language, not programming languages

Natural language is the primary medium of communication between humans.

- Computers, however, process data in binary (0s and 1s)

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How can we make machines understand human language?

NLP tasks and applications

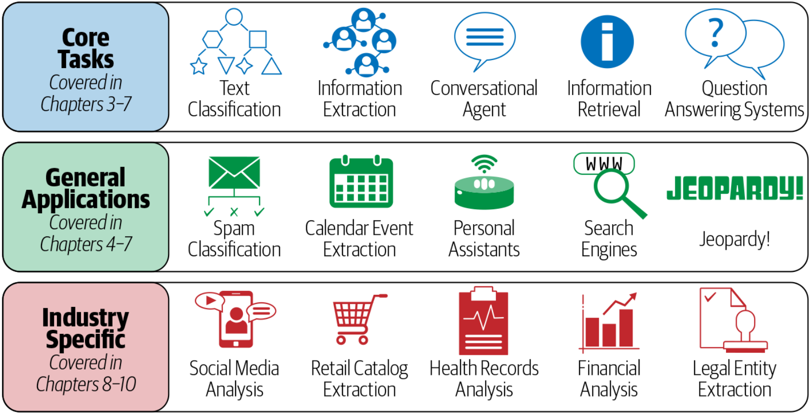

Figure: NLP tasks and applications

Why Is NLP Challenging ?

- Ambiguity, I made her duck.
    - I cooked a duck for her.
    - I made her bend down to avoid an object.
- Common knowledge,
    - Dog bit man.
    - Man bit dog.
- Creativity: Various styles, dialects, etc.
- Diversity across languages: French, English, etc.

Information extraction

- **Named Entity Recognition (NER):** Identifying names of people, places, organizations, etc.
- **Part-of-Speech (POS) tagging:** Assigning grammatical categories to each word.
- **Keyword Extraction:** Detecting key terms or phrases that summarize the text.
- **Sentiment Analysis:** Assessing the emotional tone of the text.

Applications:

- Information Retrieval
- Summarization
- Question Answering Systems

Generic NLP Pipeline

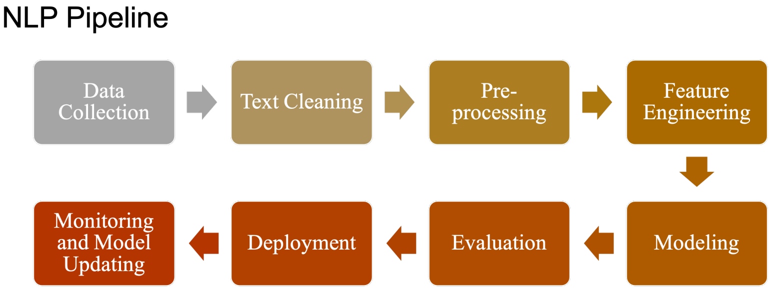

Figure: NLP Pipeline

Pre-processing

1. **Sentence Segmentation**:
   Input: "The sun is shining. It's a beautiful day."
   Output: ["The sun is shining.", "It's a beautiful day."]
2. **Word Tokenization**:
   Input: "The sun is shining."
   Output: ["The", "sun", "is", "shining"]
3. **Stop Words Removal**:
   Input: ["The", "sun", "is", "shining"]
   Output: ["sun", "shining"]
4. **Stemming / Lemmatization**:
   Input: ["running", "jumps"]
   Output after stemming: ["run", "jump"]
   Output after lemmatization: ["run", "jump"]

Stemming / Lemmatization (1)

Differences:

- **Stemming:**
  Reduces words to their base or root form by removing suffixes or prefixes, often without regard for whether the root word is valid. It uses heuristic rules (e.g., "running" becomes "run," "happily" becomes "happi").
- **Lemmatization:**
  More sophisticated process that reduces words to their base form (lemma) while considering the context and meaning of the word. It requires part-of-speech tagging (e.g., "running" becomes "run," but "better" becomes "good").

Stemming / Lemmatization (2)

Challenges:

- **Stemming:**
  - Often leads to incorrect base forms (e.g., "studies" becomes "studi").
  - Over-stemming or under-stemming can cause issues in text normalization.
- **Lemmatization:**
  - Computationally more expensive as it requires contextual understanding and part-of-speech tagging.
  - Depends heavily on accurate linguistic databases, which may not always be available for all languages or domain-specific jargon.

Feature Engineering

Definition

Feature Engineering refers to the set of the methods that feeds the pre-processed text into the ML models.

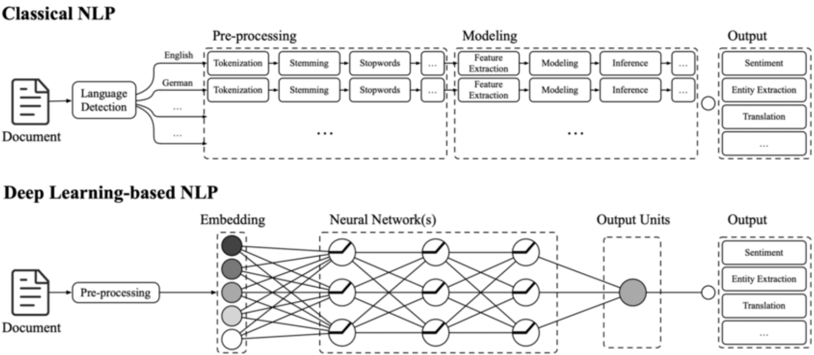

Figure: Feature engineering for classical NLP versus DL-Based NLP

Text Representation Methods (1)

- **One-Hot Encoding**:
  Vocabulary: ["sun", "is", "shining"]
  Sentence: "sun is shining"
  One-Hot Encoded Vectors:
  sun: [1, 0, 0]
  is: [0, 1, 0]
  shining: [0, 0, 1]

- **Bag of Words**:
  Sentences: ["The sun is shining", "The sun is bright"]
  Vocabulary: ["sun", "is", "shining", "bright"]
  Bag of Words Representation:
  Sentence 1: [1, 1, 1, 0]
  Sentence 2: [1, 1, 0, 1]

Text Representation Methods (2)

- **Bag of N-Grams (bigrams = 2-Grams)**:
  Sentence: "The sun is shining"
  Bigrams: ["The sun", "sun is", "is shining"]
  Bag of Bigrams Representation:
  The sun": 1, "sun is": 1, "is shining": 1

- **TF-IDF (Term Frequency-Inverse Document Frequency)**:
  Documents: ["The sun is shining", "The sun is bright"]
  Vocabulary: ["sun", "is", "shining", "bright"]
  TF-IDF Values (for the first sentence):
  sun: 0.5, is: 0.5, shining: 0.7, bright: 0

TF-IDF (Term Frequency-Inverse Document Frequency)

$$\text{TF-IDF}(t, d, D) = \text{TF}(t, d) \times \text{IDF}(t, D) \tag{1}$$

where:

$$\text{TF}(t, d) = \frac{\text{count of t in d}}{\text{number of words in d}} \tag{2}$$

$$\text{DF}(t) = \text{occurrence of t in documents}$$

(3)

$$\text{IDF}(t) = \log \frac{N}{DF(t)} \tag{4}$$

Topic Modeling (1)

Definition: Topic modeling is an unsupervised ML technique used to identify hidden semantic structures. It helps uncover the underlying topics that are prevalent in a corpus into those topics.

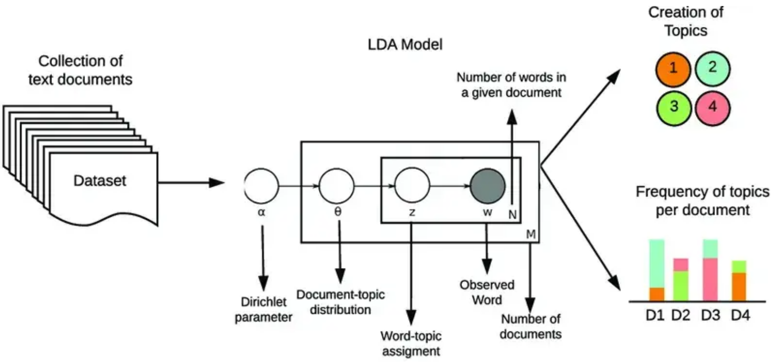

Figure: Implementation of Topic modeling methods

Topic Modeling (2)

Common Methods:

- **Latent Dirichlet Allocation (LDA):**
  A probabilistic model that assumes each document is a mixture of topics, and each topic is a mixture of words.
- **Non-Negative Matrix Factorization (NMF):**
  A matrix factorization method that decomposes the term-document matrix into topics and words, assuming non-negative weights.
- **Latent Semantic Analysis (LSA):**
  A linear algebra-based method that uses singular value decomposition (SVD) to reduce the dimensionality of the term-document matrix to discover topics.
- **Hierarchical Dirichlet Process (HDP):**
  An extension of LDA that allows the number of topics to be determined automatically from the data.

Sentiment Analysis (1)

Definition:
Sentiment analysis, also known as opinion mining, is a NLP technique used to determine the emotional tone behind a body of text. (feedback, reviews, or social media posts, etc)

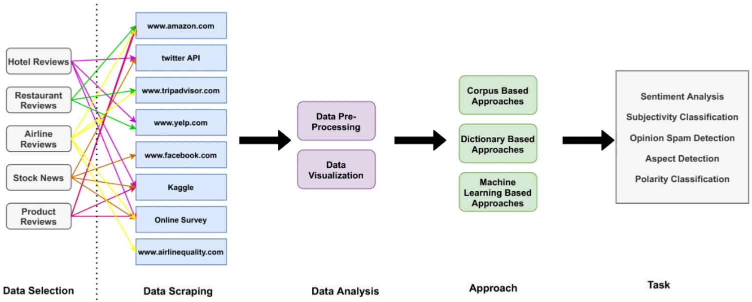

Figure: Sentiment Analysis Methodology

Sentiment Analysis (2)

Common Methods:

- **Lexicon-Based Approach:**
  A dictionary of words with pre-assigned sentiment scores (positive, negative, neutral) is used to assess the sentiment of a text based on word occurrence.
- **Machine Learning-Based Approach:**
  A supervised learning method where a model is trained on labeled datasets (positive/negative) to classify unseen text. Common algorithms include Naive Bayes, Support Vector Machines (SVM), and Logistic Regression.
- **Deep Learning Approach:**
  Neural networks, such as recurrent neural networks (RNN) or transformers (like BERT), are used to capture the context of text and generate more accurate sentiment classifications.

Popular Python Libraries for NLP

- **NLTK (Natural Language Toolkit):**
  for tokenization, parsing, stemming, tagging, and text classification.
- **spaCy:**
  provides pre-trained models for tasks like part-of-speech tagging, named entity recognition, dependency parsing, and more. SpaCy is optimized for performance on large datasets.
  Example: Named entity recognition, dependency parsing.
- **TextBlob:**
  for sentiment analysis, noun phrase extraction, and language translation.
- **Gensim:**
  A library focused on topic modeling and document similarity analysis. It provides implementations for popular algorithms like Latent Dirichlet Allocation (LDA) and Word2Vec.